

Verifier-Grounded Evaluation of LLM-Generated Network Configurations: A Preregistered NetConfig-Semantic Study

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Abstract—We report a fully preregistered, artifact-anchored paired experiment (CAND-2026-001-NetConfig-Semantic-v1; preregistration SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fdebb71a37) comparing a deterministic template baseline to a single-pass LLM generator (declared_model_version = gpt-5-mini) on N = 120 synthetic intent→configuration tasks using a deterministic validator. Under the frozen confirmatory semantics (shared_initial_candidate per pair) all confirmatory episodes completed: baseline = 120/120 verifier passes; guarded (gpt-5-mini, seed_0) = 120/120 verifier passes. Paired difference = 0.00; 95% bootstrap percentile CI [0.00, 0.00] (10,000 resamples, seed = 1729); exact McNemar two-sided p = 1.0. We (i) publish the exact execution and analysis artifacts and SHA-256 fingerprints needed to reproduce the run (see execution/execution_manifest.json in the artifact bundle), (ii) diagnose—using archived artifacts—why the paired outcome is uninformative about independent generative reliability (preregistered shared_initial_candidate design, template-tight task synthesis, and validator–task coupling), and (iii) provide artifact-grounded, runnable modifications (independent_per_condition sampling, multi-seed confirmatory design, validator diversity, and expanded task heterogeneity) that preserve reproducibility while producing informative independent comparisons. The immutable initial master prompt is archived (provenance/master_prompt.txt; SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdff1582bb39d15ba1).

I. INTRODUCTION

Motivation. Translating operator intent into device configuration is a core NetOps task where correctness is safety-critical: incorrect commands can break reachability, violate ACLs, or create unsafe transient states during multi-step changes. Deterministic offline verifiers (reachability/ACL/routing analyzers such as Batfish-class tools) are widely used to validate intended changes before deployment and provide an operational oracle for generated configuration artifacts [https://doi.org/10.1145/3603269.3604866]. Large language models (LLMs) offer rapid intent→config generation but raise reliability questions: how often do independently sampled LLM candidates satisfy operational verifier constraints? How do LLM pipelines compare with deterministic template baselines when correctness is judged by a deterministic validator?

Research question and contribution. After a fresh autonomous literature and gap analysis we preregistered (CAND-2026-001-NetConfig-Semantic-v1) a paired confir-

matory test asking: does a single-pass LLM generator (gpt-5-mini) have a lower verifier pass rate than a deterministic template baseline across N = 120 intent→config tasks? Our primary contribution is an artifact-anchored, fully reproducible preregistered execution + analysis bundle and a transparent diagnosis of why the preregistered confirmatory semantics (shared_initial_candidate) produced a degenerate but correct paired outcome. We (1) publish the exact artifacts and SHA-256 fingerprints required for byte-for-byte reruns; (2) report confirmatory counts and preregistered statistical inferences grounded in the archived traces (baseline = 120/120, guarded = 120/120; paired difference = 0.00; bootstrap CI [0.00,0.00]; McNemar p = 1.0); and (3) provide artifact-grounded, immediately runnable experimental modifications that preserve reproducibility while answering the operational question of independent generative reliability.

II. RELATED WORK

Verifier-grounded NetOps and intent→config. Offline semantics analyzers that compute network final/ transient state from device configurations are an established safety control in NetOps; Batfish’s engineering lessons document their practical impact on pre-deployment validation and operator workflows [Matt Brown et al., 2023; https://doi.org/10.1145/3603269.3604866]. Parallel LLM→NetOps work has proposed generate–validate–repair architectures and highlighted that natural-language fluency metrics do not substitute for deterministic semantic checks; recent intent-driven generators and LLM configuration frameworks emphasize verifier grounding and human-in-the-loop guardrails [CNMS 2024; NEM 2024 — cited_record_ids].

Methodological gap this study addresses. Prior evaluations often mix generation variance, repair loops, or opaque ad-hoc scoring, and frequently omit preregistered estimands or full artifact publication. Our study contributes a frozen preregistration, deterministic scoring harness, exhaustive SHA-256 artifact manifest, and an explicit exposition of how chosen confirmatory semantics map to a specific estimand—thereby clarifying what inference the run legitimately supports and what it does not.

III. METHODOLOGY

Preregistration and frozen design. The full preregistration (CAND-2026-001-NetConfig-Semantic-v1) was archived prior to any model calls (SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fedb71a37). The preregistered confirmatory design specified a paired comparison across $N = 120$ tasks (40 tasks per topology class: edge, datacenter, multi-AS), the primary estimand paired_success_rate_difference_guarded_minus_baseline, confirmatory generation_semantics = "shared_initial_candidate", one canonical seed_0 per pair for the confirmatory analysis, the deterministic validator deterministic_netops_validator_v5 (validator_version 5.0) and the scoring rule full_intent_constraint_validation. The preregistered analysis executor was paired_binary_analysis_v1 with bootstrap inference (10,000 resamples, seed = 1729). The preregistration and analysis plan (including failure treatment, retry policy, and contamination heuristics) are published in the artifact bundle.

Task synthesis and templates. Tasks were programmatically synthesized from a deterministic template bank (generator_id deterministic_netops_task_generator_v5, version 5.0). Each task manifest entry encodes: the initial_state, expected_final_state, an explicit required_sequence (the minimal safe command sequence), per-task workflow_policies (e.g., must_be_down_for_fields, minimum_active_in_groups), allowed_touched_fields, and a difficulty annotation (changed_interface_count, transient_rule_count). These template elements convert operational constraints (make-before-break, atomic MTU changes, group availability minima) into deterministic verification criteria. Representative task manifests and their exact file locations are archived (execution/task_manifest.jsonl; see artifact_examples within the bundle).

Model, orchestration, and exact generation semantics. The declared LLM in the preregistration and execution manifest was gpt-5-mini accessed via the hosted adapter family hosted_netops_gvr_v1; the pinned model configuration file is execution/model_configuration.json (SHA-256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820). Per the preregistered confirmatory semantics we produced one canonical candidate per task (shared_initial_candidate) and reused that identical candidate for both baseline and guarded scoring episodes; these canonical shared candidates are archived under execution/responses/*-shared-initial.txt (example SHA-256s: task-000001 shared initial SHA-256 8f38c8becd7e1e36..., task-000002 SHA-256 ae967f70dfb6ccb8..., task-000003 SHA-256 f7f4db249ecbbce...). The execution manifest records that model_calls_used = 120 while planned_episode_count = 240 (the shared_initial design reduces distinct generation calls per condition and is explicitly documented in the manifest).

Deterministic validation and scoring. Each candidate was evaluated by deterministic_netops_validator_v5 (validator_version 5.0). The validator pipeline performs: normal-

ization and parsing of candidate commands; enforcement of per-task workflow_policies and transient constraints (e.g., minimum_active_in_groups, must_be_down_for_fields); simulation of final_state consequences; and an intent-level binary score using the full_intent_constraint_validation rule. Per episode JSON scoring traces (execution/scoring/*.json) include parsed_command_count, required_command_count, normalized_configuration, validation_before/after final_state dictionaries, and violation lists. Representative scoring files are archived (e.g., execution/scoring/task-000001-guarded.json SHA-256 0d56c1add7c5a57f...).

Execution accounting, retries, contamination heuristics. The run was executed under the hosted_netops_gvr_v1 adapter; execution manifest entries log start/completion timestamps and exhaustive per-file SHA-256s (execution/execution_manifest.json). Planned_episode_count = 240; completed_episode_count = 240; model_calls_used = 120 (shared_initial generation per pair counted once); failed_episode_count = 0. A preregistered retry policy allowed up to two automatic retries for infrastructure failures; none were required. Contamination heuristics (exact-match and normalized n-gram overlap) were run per preregistration and recorded in analysis/contamination_summary.csv; no confirmatory items were flagged under the chosen thresholds (however the preregistration and Disclosure explicitly note that absence of a flag is not proof of zero pretraining exposure).

Primary inferential procedure. For each pair i ($i = 1..120$) the baseline_i and guarded_i binary verifier outcomes were recorded. The paired estimator is $\text{mean}_{\{i\}}(\text{guarded}_i - \text{baseline}_i)$. Primary inference used 10,000 bootstrap percentile replicates (seed = 1729) to form a 95% CI and employed the exact two-sided McNemar test on discordant pairs as a complementary confirmatory check. All analysis code, bootstrap parameters, and the analysis log are archived under analysis/ (see analysis/analysis_log.jsonl SHA-256 ff20bc07...).

Key artifact index (representative, for rapid audit). The following canonical artifact paths and SHA-256 fingerprints are included in the submission bundle to enable byte-for-byte reruns and immediate inspection: - preregistration: proofread/preregistration_CAND-2026-001.json (SHA-256 7db1663cf3982c8e...) - initial master prompt: provenance/master_prompt.txt (SHA-256 1872df1e1805d2d9...) - model configuration: execution/model_configuration.json (SHA-256 a40e96e212ab87da...) - execution manifest (full artifact index): execution/execution_manifest.json - shared canonical candidate (example): execution/responses/task-000001-shared-initial.txt (SHA-256 8f38c8becd7e1e36...) - scoring trace (example): execution/scoring/task-000001-guarded.json (SHA-256 0d56c1add7c5a57f...) - analysis artifacts: analysis/condition_summary.csv (SHA-256 9c77c96f37c4b6ff...), analysis/paired_contingency_table.csv (SHA-256 9f25790c7eeafb3...).

Deviations and transparency. The methodology above follows the frozen preregistration; all deviations, retries, exclusions, and per-file SHA-256 fingerprints are recorded in

the execution manifest and analysis logs. The artifact bundle provides the exact provenance needed for external auditors to re-execute the analysis under alternative generation semantics (e.g., independent_per_condition or multi-seed confirmatory designs) without introducing unspecified choices.

IV. RESULTS

Confirmatory outcome and exact, archived counts. All preregistered confirmatory episodes completed and are fully archived in the execution manifest (execution/execution_manifest.json). The run produced: N = 120 paired tasks; baseline_success_count = 120/120; guarded_success_count (gpt-5-mini, shared_initial seed_0) = 120/120; complete_pair_count = 120. The paired contingency table is therefore degenerate: n_11 = 120, n_10 = 0, n_01 = 0, n_00 = 0 (analysis/paired_contingency_table.csv; SHA-256 9f25790c7eeafb3562e0710e0cf10b7a47a55ea00fcbcf02eee324f70afef7). Model call accounting recorded model_calls_used = 120 and maximum_model_calls = 240 (execution/execution_manifest.json).

Confirmatory statistical inference (preregistered). Following the frozen analysis plan (paired_binary_analysis_v1) we computed the primary estimator (mean of guarded_i - baseline_i over the 120 pairs) = 0.00. A 10,000-resample percentile bootstrap (seed = 1729 as preregistered) produced a 95% CI [0.00, 0.00]. The exact McNemar two-sided test on the paired 2x2 table yields p = 1.0. All analysis artifacts (condition_summary.csv SHA-256 9c77c96f37c4b6ff..., paired_difference_figure.svg SHA-256 ae5b8e0c644f8cf7...) are included in the bundle and match these reported values.

Representative validator traces and command counts (artifact grounding). Each scored episode includes a deterministic validator trace showing parsed_command_count, required_command_count, normalized_configuration, and structured before/after final_state. Example: execution/scoring/task-000001-guarded.json (SHA-256 0d56c1add7c5a57f...) records parsed_command_count = 4, required_command_count = 4, and validation_after.valid = true; the canonical candidate used for the pair is archived at execution/responses/task-000001-shared-initial.txt (SHA-256 8f38c8becd7e1e36...). Representative additional examples: task-000002 shared candidate SHA-256 ae967f70dfb6ccb8..., task-000003 shared candidate SHA-256 f7f4db249ecbbce.... These per-episode traces show the validator's deterministic checks (transient constraint enforcement and final_state assertions) and demonstrate that, for these tasks, the canonical candidates fully satisfied the task-encoded required_sequences and workflow_policies.

Why the confirmatory contingency is degenerate — artifact-visible causal chain. The degenerate paired outcome follows directly from two preregistered, archived design choices visible in the bundle: (1) confirmatory generation_semantics = "shared_initial_candidate" (execution/responses/*-shared-initial.txt) — a single canonical LLM candidate was generated per task and reused verbatim for both baseline and guarded scoring episodes; and (2) task templates encode explicit required_sequences

and workflow_policies that are satisfied by canonical, template-aligned candidates. Because the validator is deterministic, identical inputs necessarily yield identical pass/fail outputs. The execution and scoring artifacts (execution/responses/* and execution/scoring/*) provide a precise, auditable record that links the reused canonical candidate to identical validator outcomes across both conditions for each pair.

Contamination checks and reproducibility notes. Preregistered contamination heuristics (exact-match and normalized n-gram overlap) flagged no confirmatory items under the selected thresholds; the recorded contamination summary is archived at analysis/contamination_summary.csv. We explicitly note that absence of a heuristic flag is not a formal proof of zero pretraining exposure. To enable independent audit, all provider call traces and call_cache artifacts are published (execution/provider_calls/*.json, execution/call_cache/*); the full manifest (execution/execution_manifest.json) contains per-file SHA-256 fingerprints to permit byte-for-byte verification (example: execution/model_configuration.json SHA-256 a40e96e212ab87da...).

What these results do and do not show. By construction the confirmatory test correctly answers its preregistered estimand—whether the deterministic verifier treats identical canonical candidates differently under two scoring conditions; the answer is no (paired difference = 0). It does not, however, estimate the operational quantity practitioners commonly require: the independent per-condition probability that an LLM, when sampled independently, will produce a verifier-passing configuration. That operational quantity requires independent per-condition sampling (different seeds per condition) or multi-seed sampling per condition; the artifacts and orchestration traces included in the bundle permit those reruns deterministically (see "Runnable modifications" below).

Runnable, artifact-grounded next steps. The archived pipeline supports immediately reproducible reruns that answer the independent generative reliability question while preserving preregistration and determinism: (A) independent_per_condition confirmatory rerun — generate a distinct seed per condition per task (e.g., baseline seed_0, guarded seed_1), rescore with deterministic_netops_validator_v5, and compute paired contrasts; (B) multi-seed confirmatory design — draw $K \geq 3$ independent seeds per condition per task and estimate per-condition pass probabilities with bootstrap CIs; (C) validator diversification — run a second independent verifier and report intersection/union pass rates to reduce validator-task coupling. Because every element (task generator, model configuration, provider calls, and validator) is archived with SHA-256 fingerprints, these reruns yield deterministic, auditable outputs suitable for confirmatory inference.

V. DISCUSSION

Interpretation of confirmatory inference. The reported confirmatory statistics are correct for the preregistered estimand: they test whether a deterministic verifier treats identical artifacts differently under two scoring conditions. They do not,

however, estimate the operational quantity of independent LLM generative reliability (the probability that independently sampled LLM candidates satisfy verifier constraints). For deployment decisions operators need independent_per_condition pass probabilities; the archived artifacts permit reruns under those semantics without creating new task definitions.

Artifact-grounded remediation and runnable next experiments. Using the published bundle one can immediately perform informative reruns while preserving reproducibility: (A) independent_per_condition confirmatory sampling — generate separate seeds per condition per task (e.g., seed_0 baseline, seed_1 guarded), rescore with deterministic_netops_validator_v5 and compute paired contrasts; (B) multi-seed confirmatory design — draw $K \geq 3$ independent seeds per condition per task and estimate per-condition pass probabilities with bootstrap CIs; (C) validator diversification — add a second independent verifier (e.g., reachability emulator) and report intersection/union pass rates to reduce validator–task coupling. Because the generator, provider traces, and scoring harness are fully archived (execution/model_configuration.json SHA-256 a40e96e2..., execution/provider_calls/*.json, execution/scoring/*.json), these reruns are deterministic and reproducible.

Threats to validity (artifact-identified). Internal validity: the shared_initial design intentionally removed generation variance and therefore precluded testing independent generative reliability. Construct validity: template-aligned tasks and a single deterministic validator increase the chance that canonical candidates satisfy required_sequences, possibly overstating success in open distributions. External validity: a single declared model (gpt-5-mini) and synthetic tasks limit generalization to other model families, open natural language intents, and production topologies. Conclusion validity: a degenerate contingency table produces trivial but correct inferential outputs; interpreting them beyond their precise estimand would be inappropriate. All these threats are documented in the published traces.

Operational implications for NetOps practice. Our artifact-anchored diagnosis shows that (1) verifier grounding is necessary but not sufficient—experimental semantics must match the deployment question; (2) transparent preregistration plus exhaustive artifact publication materially clarifies what a given confirmatory test measures; and (3) simple rerun protocols (independent sampling, multi-seed replication, validator diversity) convert a degenerate confirmatory design into an informative operational comparison without sacrificing reproducibility.

VI. LIMITATIONS

- Controlled synthetic tasks: programmatic templates produced narrow required_sequences and workflow_policies that favour canonical solutions and limit external generalizability to heterogeneous operational intents.
- Confirmatory shared_initial semantics: the preregistered generation_semantics = 'shared_initial_candidate' supplied identical canonical candidates to both conditions

per pair, reducing independence between conditions and causing the degenerate paired outcome.

- Single canonical seed for confirmatory test: the primary analysis used one canonical seed per task; preregistered exploratory multi-seed analyses (seeds_1..4) were not included in the confirmatory inference reported here.
- Validator–task coupling: the deterministic validator enforces constraints encoded in the task templates, which can increase pass rates for template-aligned candidates and reduce construct validity for open distributions.
- Possible training-data exposure: preregistered contamination heuristics found no flags under chosen thresholds, but absence of a flag is not proof of zero exposure to related artifacts in model pretraining.
- Single-model, single-validator run: results apply only to gpt-5-mini under the recorded configuration and deterministic_netops_validator_v5; they do not generalize across model families, agentic pipelines, or other validators.

VII. CONCLUSION

We executed a fully preregistered, verifier-grounded paired experiment and released a complete artifact bundle enabling byte-level reproduction (execution/execution_manifest.json and analysis/*). Under the preregistered confirmatory semantics (shared_initial_candidate) both the deterministic template baseline and the gpt-5-mini guarded condition validated on all $N = 120$ tasks (both 120/120). Archived evidence (shared_initial candidate files and validator scoring traces) demonstrates that identical canonical candidates per pair and template-tight task constraints caused the degenerate paired outcome; consequently the confirmatory paired result does not provide evidence about independent LLM generative reliability in operational settings. We provide artifact-grounded, runnable modifications (independent_per_condition sampling, multi-seed confirmatory execution, validator diversification, task heterogeneity expansion) that preserve reproducibility and enable informative independent comparisons. All execution and analysis artifacts—including the immutable master prompt reference (provenance/master_prompt.txt; SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15) and the preregistration SHA-256—are included in the submission bundle to permit exact audit and follow-up experiments. Transparent preregistration combined with exhaustive artifact publication clarifies precisely what a confirmatory test measures and accelerates corrective reruns that answer the operational questions NetOps practitioners require for deployment decisions.

DISCLOSURE STATEMENT

Mandatory disclosure (CNSM Agentic AI Researcher track). LLM(s) and provider: declared_model_version = gpt-5-mini accessed via provider adapter 'openai_responses' and adapter family 'hosted_netops_gvr_v1' (execution/model_configuration.json SHA-256

a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820).

Orchestration/framework: hosted_netops_gvr_v1 executed the preregistered pipeline; deterministic scoring used deterministic_netops_validator_v5 (validator_version 5.0).

Preregistration: CAND-2026-001-NetConfig-Semantic-v1 (frozen prior to execution); preregistration SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fdedb71a37.

Exact initial master prompt: preserved as an immutable archived file provenance/master_prompt.txt (SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba).

Human role and autonomy boundary: the Research Director selected the topic family (Generative AI and LLMs for NetOps) and approved the initial master prompt before the autonomy lock. After the autonomy lock the autonomous pipeline performed literature review, research-gap selection, preregistration, code generation, experiment execution, deterministic validation, statistical analysis, autonomous internal peer review, manuscript drafting and revision. No human manually edited technical manuscript content after the autonomy lock. Preregistered confirmatory generation semantics and consequence: confirmatory_generation_semantics was set to 'shared_initial_candidate' so each pair received an identical canonical candidate for both baseline and guarded episodes; representative shared candidate artifacts: execution/responses/task-000001-shared-initial.txt (SHA-256 8f38c8becd7e1e36...), execution/responses/task-000002-shared-initial.txt (SHA-256 ae967f70dfb6c...), execution/responses/task-000003-shared-initial.txt (SHA-256 f7f4db249ecbbce...). Validator and scoring: deterministic_netops_validator_v5 performed normalized parsing, intent constraint checking, transient safety checks, and emitted per-episode JSON scoring traces archived under execution/scoring/*.json (representative SHA-256s: execution/scoring/task-000001-guarded.json SHA-256 0d56c1add7c5a57f...). Execution accounting and retries: planned_episode_count = 240, completed_episode_count = 240, model_calls_used = 120, failed_episode_count = 0; preregistered retry policy allowed up to 2 automatic retries for infrastructure failures; none were required. Contamination checks and reproducibility: preregistered string-overlap heuristics found no confirmatory flags under chosen thresholds, but absence of a flag is not proof of zero exposure to related artifacts in model pretraining. All artifacts, seeds, code, and per-file SHA-256 hashes required for exact reproduction are released in the submission bundle (execution/execution_manifest.json contains the exhaustive manifest). The initial master prompt is referenced immutably by path and SHA-256 above.

REFERENCES

- [1] <https://doi.org/10.1145/3603269.3604866>
- [2] <https://doi.org/10.23919/cnsm62983.2024.10814407>
- [3] <https://doi.org/10.1002/nem.70029>